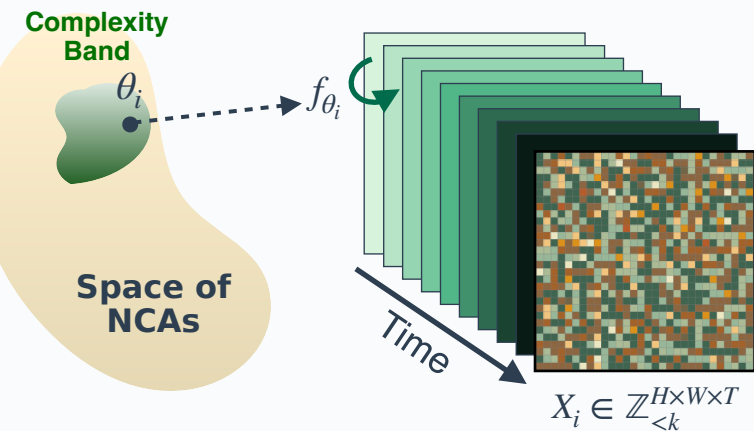
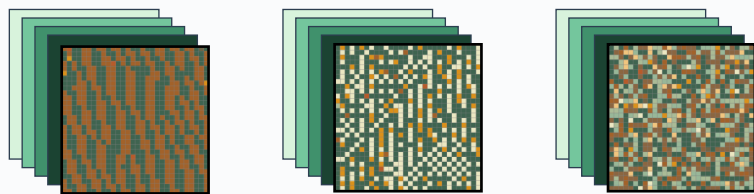


Pre-Pre-Training



LLM



Low

Medium

High

Complexity

Pre-Training

By the end of 2017, every smartphone manufacturer has embraced the 18:9 aspect ratio screen model...

Whenever I start on a large refactoring, I need to justify that the exercise...

```
from typing import List
def has_close_elements(
    numbers: List[float],
    threshold: float
) -> bool:
    """
```

```
    check if in given
    list of numbers,
    are any two are
    closer to each
    other than the
    threshold
    """ ...
```

Let $a_1, a_2, \dots, a_n$ be a sequence of positive integers such that $a_1 + a_2 + \dots + a_n = n^2$. Prove that there exist indices $i < j$ for which $a_i + a_{i+1} + \dots + a_j$ is a perfect square.

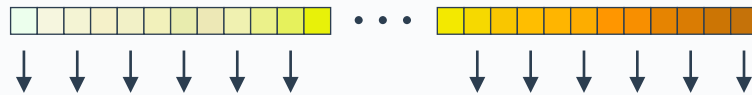
Solution: Consider the partial sums $S_k = a_1 + a_2 + \dots + a_k$

WebText

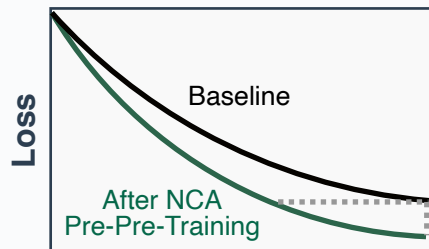
Code

Math

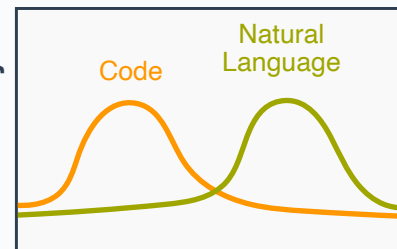
Natural Language Data



LLM



Pre-training Tokens



NCA Complexity